LungLogic: Intelligent Airflow Control for Realistic Breathing Simulation

ME-12-TYIN-007 | Yuqiao Li, Chun Yin Ku (John), Jiaquan Zhong | Supervisors: Jonathan Eden, Ying Tan

1. Introduction

1.1 Project need

Adjustable RR, I:E, tidal volume and regional airflow reproduce patient-like lung conditions for repeatable lung-imaging research.

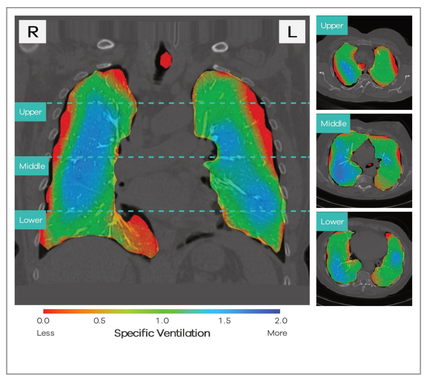


Fig. 1. XV LVAS ventilation map.

1.2 Project objective

Commercial models cannot meet specific research and development needs. 4DMedical's XV scanner enables research using our controller and the medical team's lung model. This stage validates one 250 mL branch prototype.

1.3 Nominal breathing reference

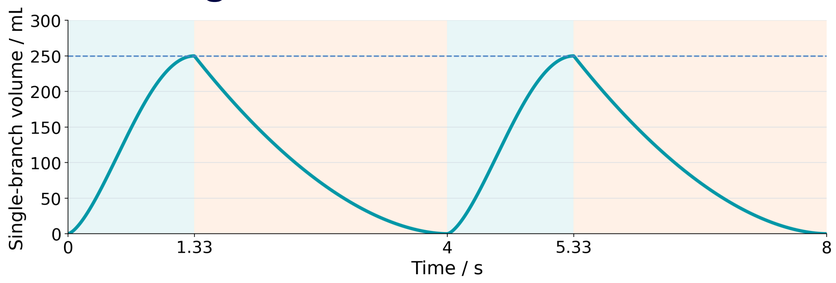


Fig. 2. Breathing-cycle illustration at 15 bpm and I:E = 1:2.

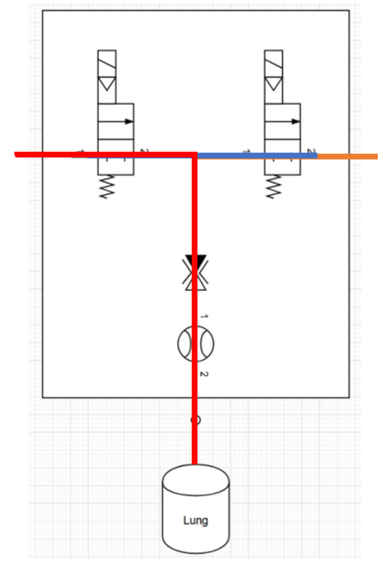
Respiratory rate	12–16 bpm; nominal 15 bpm
Inhale:Exhale time	1:2; 1.33 s / 2.67 s at 15 bpm
Tidal volume	250 mL per branch
Regional airflow	Independently programmable by branch

Nominal reference values shown; operating parameters are programmable.

Source: 4DMedical, XV LVAS Info Sheet, DOC-5070 US V1 (2023), pp. 1-2.

2. Pneumatic System & Hardware

2.1 Pneumatic system prototype



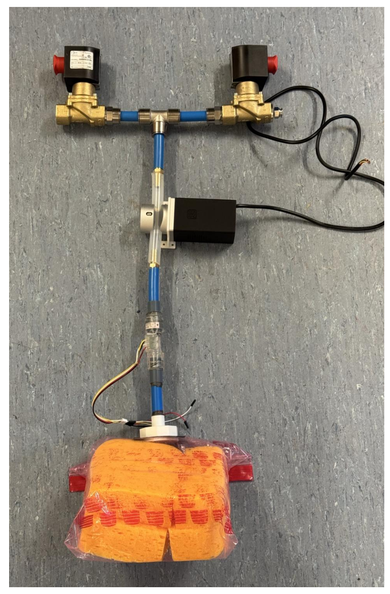


Fig. 3. Single-lung pneumatic route and 250 mL prototype.

2.2 Embedded control architecture

PC / Python UI → **MCU** → **24 V bridge PCB** → **Valves + sensor**

MCU executes commands from UI and acquires flow data. A custom bridging PCB integrates 24 V valve drivers, proportional-valve control circuitry, and sensor interfaces.

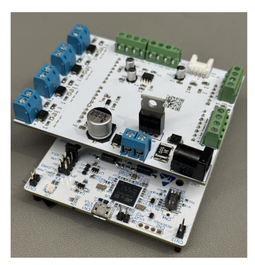


Fig. 4. Control architecture and bridge PCB connected to STM32.

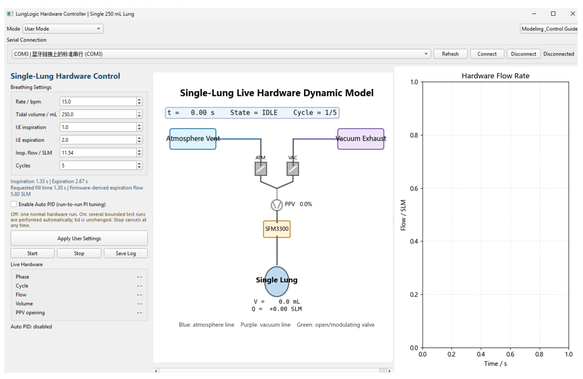
3. Integrated Control System & Software

3.1 Objective

Track flow and tidal volume with a visible, repeatable closed-loop system. Feedforward supplies the nominal valve command; sensor feedback corrects error.

3.2 Operator and developer interface

User Mode



Developer Mode

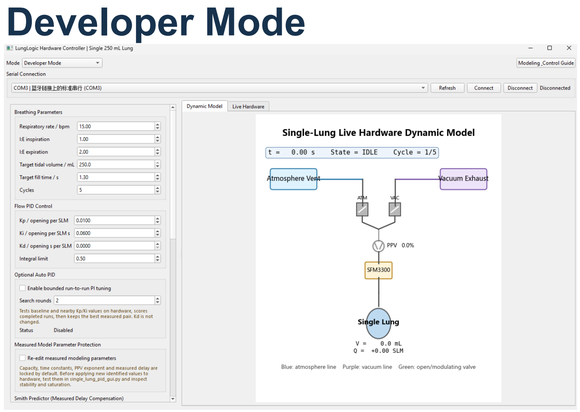


Fig. 5. User and Developer modes.

3.3 Model-assisted flow feedback

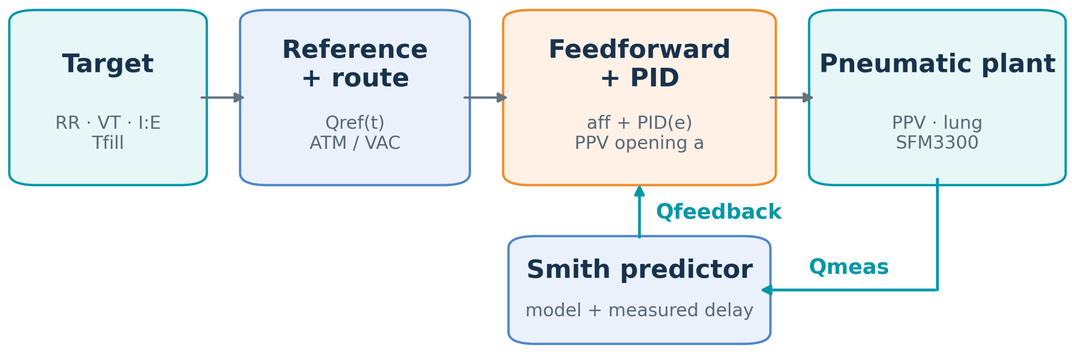


Fig. 6. Feedback control with optional Smith prediction.

3.4 Optional Auto PID and measured-delay compensation

Auto PID scores bounded Kp/Ki over full runs by flow, volume and saturation; optional Smith prediction uses the model and measured delay.

4. Results

4.1 Digital-twin simulation

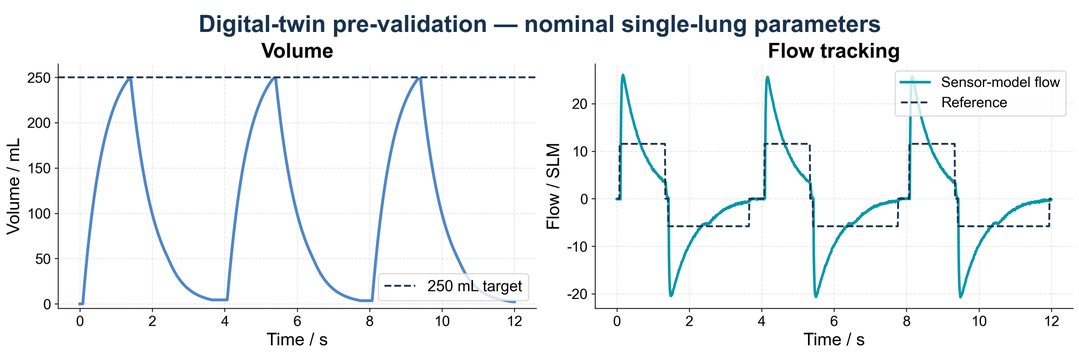


Fig. 7. Digital-twin volume and flow tracking at 15 bpm.

4.2 Physical measurements

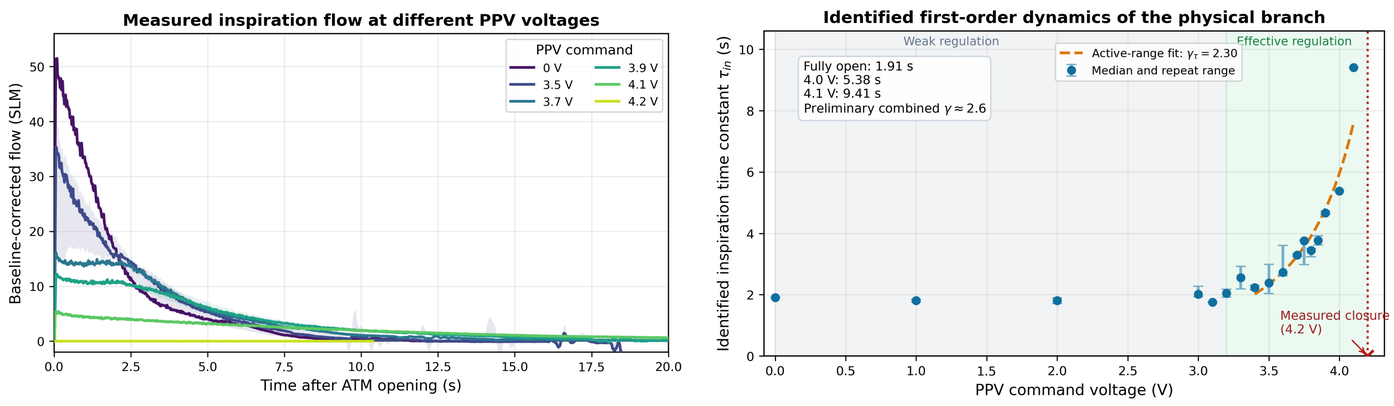


Fig. 8. Measured PPV flow response and identified first-order dynamics.

The measured PPV response qualitatively matches the simulation model: increasing command voltage attenuates inspiratory flow, and the transient follows a first-order decay.